Snowflake Cloud Data Warehouse Cheat Sheet

Markaj Agron Jorand Yuuta Stampfli Nathan Liao Pei-Wen

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1 Why Snowflake?

1.1 Real-world Scenario: SwissBike SA

SwissBike SA is a Swiss SME selling bicycles and accessories online. Data comes from multiple sources: e-commerce platform, ERP system, CRM system and marketing tools.

Management needs answers to questions such as: Which countries generate the most revenue? Who are the most valuable customers? Which products should be promoted? How are sales evolving over time?

Without a centralized solution, data remains scattered across systems, making reporting slow, inconsistent and difficult to maintain.

1.2 What does Snowflake do?

Snowflake is a cloud-native data warehouse that centralizes data from multiple sources and enables large-scale analytical processing.

1.2.1 Data Storage Characteristics

Supports structured, semi-structured and unstructured data; native support for JSON, Avro, ORC, Parquet and XML; automatic compression; micro-partitioning; separation of Storage and Compute.

1.2.2 Inputs and Outputs

Inputs
CSV files, databases, ERP systems, CRM systems, APIs, streaming data.

Outputs
SQL query results, dashboards (Power BI, Tableau, Looker), reports, machine learning and analytics data.

2 Benefits and Limitations

2.1 Benefits

Separation of Storage and Compute; broad data support; high scalability; excellent concurrency; massively parallel processing (MPP); minimal administration; strong security; Time Travel & Fail-safe; fast analytical performance; secure data sharing; multi-cloud support; automatic cost optimization.

2.2 Limitations

Cost; unpredictable spending; vendor lock-in; cloud-only platform; limited infrastructure control; Time Travel storage costs; learning curve; not optimized for OLTP workloads; dependency on cloud providers.

3 Cost Structure

3.1 Storage

Measured in TB stored per month.

```
1 TB stored data
Cost: approximately $23/month
```

3.2 Compute

Measured in credits consumed by Virtual Warehouses. Cost depends on warehouse size, execution time and number of clusters.

4 Monthly Cost Example

4.1 SwissBike SA

Assumptions: 1 TB storage, 10 business users, 10 hours warehouse usage/week, X-Small warehouse, Snowflake AI features enabled.

Component	Cost
Compute	$117.00
CoWork	$123.08
CoCo	$5.78
Storage	$23.00
Total	$268.86

Observations: Compute is the largest cost component. AI features significantly increase monthly spending. Storage costs remain relatively low. Automatic warehouse suspension helps reduce expenses.

5 Getting Started

5.1 Prerequisites

• Snowflake account • Access to AWS, Azure or GCP • Warehouse creation permissions

5.2 Setup Steps

1. Create database → 2. Create schema → 3. Create warehouse → 4. Load data → 5. Execute queries

5.3 Basic SQL Operations

Snowflake relies on standard SQL for data definition (DDL), data manipulation (DML) and administration tasks.

```
CREATE DATABASE POC_SNOWFLAKE;
CREATE SCHEMA DEMO;

CREATE WAREHOUSE DEMO_WH
WITH WAREHOUSE_SIZE='XSMALL';

SELECT CURRENT_ACCOUNT(),
       CURRENT_USER(),
       CURRENT_WAREHOUSE();

SELECT *
FROM SNOWFLAKE_SAMPLE_DATA
.TPCH_SF1.CUSTOMER
LIMIT 10;
```

6 Common Commands

Query sample data

```
SELECT *
FROM SNOWFLAKE_SAMPLE_DATA
.TPCH_SF1.ORDERS
LIMIT 100;
```

Warehouse operations

```
ALTER WAREHOUSE COMPUTE_WH
SET WAREHOUSE_SIZE='SMALL';

ALTER WAREHOUSE COMPUTE_WH
SUSPEND;

ALTER WAREHOUSE COMPUTE_WH
RESUME;
```

Time Travel

```
DROP TABLE CUSTOMER_PROD;
UNDROP TABLE CUSTOMER_PROD;
```

Time Travel recovers historical versions of data; Fail-safe provides an additional recovery period.

Zero-Copy Clone

```
CREATE TABLE CUSTOMER_TEST
CLONE CUSTOMER_PROD;
```

Query Profile

Monitoring → Query History → Query Profile.

Displays Table Scan, Join, Aggregate and Sort operations, helping understand MPP execution.

7 When Should a SME Use Snowflake?

Recommended: Business Intelligence, Data Warehousing, Analytics, Reporting, Data Science, Multi-source Data Integration.

Not Recommended: Small applications with limited data, OLTP workloads, projects requiring full infrastructure control.

Recommendation: Snowflake is an excellent choice for SMEs requiring scalable analytics with minimal operational overhead, provided costs and vendor lock-in risks are actively monitored.